

# CSE351 Computer Networks

## Programming Assignment #2 Report

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I implemented the HTTP proxy in C using basic socket programming and process forking. The program listens on a user-specified port and forwards HTTP/1.0 GET requests between clients and servers.

To support **multiple clients**, I used `fork()` after each `accept()` so every client connection runs in a separate process. This allows concurrent requests without blocking the main server loop. I also installed a `SIGCHLD` handler to reap terminated child processes.

For **HTTP request parsing**, I wrote a simple parser that checks the request line, extracts the host, port, and path, and ensures that only valid `GET` and `HTTP/1.0` requests are handled. If the format is invalid, the proxy returns a 400 Bad Request response.

I added a **basic caching mechanism** to improve performance. Cached responses are stored as files in a `./cache` directory, with filenames generated from a hash of the URL. Each cache entry stores its expiration time based on the `Cache-Control` header. When a request matches a valid cached entry, the proxy serves the file directly without contacting the remote server.

The proxy reads and forwards data in chunks using a buffer of 8192 bytes. It inspects the HTTP headers once to determine caching policy, then streams the data to the client while optionally storing it to the cache.

For safety and portability, I made sure the code creates the cache directory if missing, ignores unsupported methods, and uses `SO_REUSEADDR` to avoid port conflicts.

By default, the proxy produces no console output, matching the grading requirements. The implementation supports multiple clients, handles caching with expiration, and works with standard HTTP/1.0 clients such as telnet or browsers.

Screenshot of passing all the tests from `proxy_tester/proxy_tester`:

```
113 |         if (len >= sizeof(line)) len = sizeof(line) - 1;
    |         ^
proxy.c: In function 'parse_request':
proxy.c:285:21: warning: comparison of integer expressions of different signedness: 'int' and 'long unsigned int' [-Wsign-compare]
   285 |         if (len >= sizeof(url_host_part)) len = sizeof(url_host_part) - 1;
    |         ^
proxy.c: In function 'check_cache':
proxy.c:66:21: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
    66 |         fread(&content, 1, &size, fp);
    |         ^
proxy.c: In function 'save_to_cache':
proxy.c:98:9: warning: '_builtin_strncpy' output may be truncated copying 255 bytes from a string of length 255 [-Wstringop-truncation]
    98 |         strncpy(cache_table[idx].filename, filename, sizeof(cache_table[idx].filename) - 1);
    |         ^
gcc -Wall -Wextra -O2 -std=gnu99 -o proxy proxy.o
root@e5bc25e86775:/workspace#
root@e5bc25e86775:/workspace# Display all 711 possibilities? (y or n)
root@e5bc25e86775:/workspace# ls
Dockerfile  Makefile  cache  proxy  proxy.c  proxy.o  proxy_tester
root@e5bc25e86775:/workspace# make
make: Nothing to be done for 'all'.
root@e5bc25e86775:/workspace# $ chmod +x ~/proxy_tester/proxy_tester
bash: $: command not found
root@e5bc25e86775:/workspace# make
make: Nothing to be done for 'all'.
root@e5bc25e86775:/workspace# ./proxy 5678
^C
root@e5bc25e86775:/workspace# ls
Dockerfile  Makefile  cache  proxy  proxy.c  proxy.o  proxy_tester
root@e5bc25e86775:/workspace# file proxy
bash: file: command not found
root@e5bc25e86775:/workspace# ls
Dockerfile  Makefile  cache  proxy  proxy.c  proxy.o  proxy_tester
root@e5bc25e86775:/workspace# make
make: Nothing to be done for 'all'.
root@e5bc25e86775:/workspace# file proxy
bash: file: command not found
root@e5bc25e86775:/workspace# ./proxy 5678
^C
root@e5bc25e86775:/workspace# make
make: Nothing to be done for 'all'.
root@e5bc25e86775:/workspace# ls
Dockerfile  Makefile  cache  proxy  proxy.c  proxy.o  proxy_tester
root@e5bc25e86775:/workspace# ./proxy 5678
^C
root@e5bc25e86775:/workspace# make
make: Nothing to be done for 'all'.
root@e5bc25e86775:/workspace# ls
Dockerfile  Makefile  cache  proxy  proxy.c  proxy.o  proxy_tester
root@e5bc25e86775:/workspace# ./proxy 5678
^C
root@e5bc25e86775:/workspace#
```

```
mac - root@e5bc25e86775:/workspace - docker exec -it proxy bash - 117x64
### Testing error case #2:
Request with method other than GET (case 1): [PASSED] (2/2 points)
Request with method other than GET (case 2): [PASSED] (2/2 points)
Request with method other than GET (case 3): [PASSED] (2/2 points)
Request with method other than GET (case 4): [PASSED] (2/2 points)

### Testing error case #3:
Request with different HTTP version (other than v1.0): [PASSED] (8/8 points)

### Testing error case #4:
Invalid host header: [PASSED] (8/8 points)

### Testing error case #5:
Different URI and host: [PASSED] (8/8 points)

Total Score: 65 / 68
root@e5bc25e86775:/workspace# ./proxy_tester/proxy_tester ./proxy 5678
Binary: ./proxy
Running on port 5678
#####
Basic Functionality Test
#####
### Testing: http://www.kaist.edu
http://www.kaist.edu: [PASSED] (5/5 points)

### Testing: http://info.in2p3.fr/
http://info.in2p3.fr/: [PASSED] (5/5 points)

### Testing: http://info.cern.ch/
http://info.cern.ch/: [PASSED] (5/5 points)

### Testing: http://info.cern.ch/hypertext/WWW/TheProject.html
http://info.cern.ch/hypertext/WWW/TheProject.html: [PASSED] (5/5 points)
#####
Extended Functionality Test
#####
### Testing error case #1:
Request without host header field: [PASSED] (8/8 points)

### Testing error case #2:
Request with method other than GET (case 1): [PASSED] (2/2 points)
Request with method other than GET (case 2): [PASSED] (2/2 points)
Request with method other than GET (case 3): [PASSED] (2/2 points)
Request with method other than GET (case 4): [PASSED] (2/2 points)

### Testing error case #3:
Request with different HTTP version (other than v1.0): [PASSED] (8/8 points)

### Testing error case #4:
Invalid host header: [PASSED] (8/8 points)

### Testing error case #5:
Different URI and host: [PASSED] (8/8 points)

Total Score: 68 / 68
root@e5bc25e86775:/workspace# ./proxy_tester/proxy_tester ./proxy 5678
Binary: ./proxy
```